

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>
we know the file size is 200 gb so we can write that down
then we know carla downloads at a speed of 2 gb per minute so we can work out how long it takes her to download 40 percent of the file
so we do 40 percent of 200 gb which we write as 40 percent times 200 so we do the sum 40 divided by 100 to turn the percent into a number we can work with then times that by 200
so 40 divided by 100 is 0 point 4 then we times 0 point 4 by 200 which equals 80 so now we know 40 percent of the file is 80 gb
then we know carla can download 2 gb a minute so to work out how long it takes her to download 80 gb we do 80 divided by 2 which equals 40 so now we know it takes 40 minutes to download 40 percent of the file
then the question tells us there's a 20 minute restart so we add the 40 minutes from the download to the 20 minutes from the restart so 40 plus 20 equals 60
then the question tells us after that she has to start the download from the beginning again so now we do the same sum as we did at the start for the full file size so we take 200 gb and work out how long it takes to download at 2 gb a minute so we do 200 divided by 2 which equals 100
so now we know it takes 100 minutes to download the full 200 gb
so now we add the 60 minutes from before the full 100 minutes which equals 160
so now we know it takes 160 minutes in total
</think><answer>160</answer>